

Bagging technique.

↳ group of trees

- ↳ random sampling.

- Bagging in decision tree = random forest

Q) Why Random Forest gives better answer?
Random Forest + Bias Variance.

$$\text{Bias} \propto \frac{1}{\text{Volume}}$$

Low Bias High Variance

- Full Decision Tree
- SVM
- KNN

High Bias Low Variance

- Linear Reg
- Logistic Regression -

Decision Tree: LBHV

- splits dataset, splits noisy data and distribute.

Hence low variance.

Hence; $\angle B, \angle V$


```
from sklearn.ensemble import RandomForestClassifier
rf = RandomForestClassifier(n_estimators = 500,
random_state = 42)
```

Decision Tree Regressor

Bagging Vs Random Forest:

Random Forest
↓
DT

Bagging
↓
DT KNN SUM

⑥ If we do bagging by using Decision Tree; then does it Random Forest?

→ NO:

Reason → Feature Selection / Feature sampling.

↓
row column

In bagging:

Let say there are 5 columns and doing column sampling: taking 2 colⁿ.

① 2 ③ 4 5

- In bagging:

colⁿ will be

sampled only

once

let say 1 & 3

either in 1

or 3 in each

node and branch.

- tree level sampling

Random forest

- But in ~~decision tree~~:

colⁿ will be sampled

in each node

1 3

called

node level

in each node

sampling

diff sampling will be done.

1 5

2 4

more randomness & better

• bag = Bagging Classifier (max-features = 2)

rf = Random Forest Classifier (max-features = 2)

OOB Score

sampling with replacement:

- some repeats, but some rows remain unused at least once, also these are known as out of Bag sample.

- These aren't select to go in decision tree.

And, these points aren't seen by our model.

- we can use them for testing.

$\sim 37\%$ rows remain unused [OOB sample]

of oob-score